



DATA STRUCTURE AND ALGORITHM

Programming Practice Session

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Instructor Information

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Regulations and Grading Policy

- ❖ 6 practical sessions and 1 practical exam.
- ❖ Sign the attendance sheet in each session.
- ❖ Turn off or switch your phone to silent mode in class.
- ❖ Avoid being late or absent from class.
- ❖ Emails or messages requesting grades will not be answered.
- ❖ Grades reflect the outcomes of the learning process.
- ❖ Check email, courses, group regularly.



Regulations and Grading Policy

- ❖ The practical session accounts for **30%** of total grade.
- ❖ Each session includes: attendance test, end-of-class test and homework assignments.
- ❖ The practical exam accounts for **70%** of the practical.
- ❖ Bonus points (up to 1 point) will be added to the practical exam score and/or may determine the final rounding of the grade.
- ❖ C++ required.



Regulations and Grading Policy

- ❖ Code must include comments
- ❖ Use appropriate data types and meaningful names for variables and functions.
- ❖ Submit assignments according to the requirements, with filename format: AssignmentOrder-StudentID-FullName, e.g., “[Lab1]23529999-NguyenVanA.cpp”
- ❖ Paper tests must include: Full Name, Student ID and Session.



1st Session

Big-O Notation (Complexity)

Searching

Sorting



Quiz

1. Which of the following is a linear data structure?

A. Array

B. Linked List

C. Binary Tree

D. Graph



Quiz

2. What is the worst-case time complexity of Linear Search?

A. $O(1)$

B. $O(\log n)$

C. $O(n)$

D. $O(n \log n)$

Quiz



3. Which sorting algorithm is NOT comparison-based?

A. Merge Sort

B. Quick Sort

C. Counting Sort

D. Heap Sort

Quiz



4. What is the average-case time complexity of Binary Search?

A. $O(1)$

B. $O(\log n)$

C. $O(n)$

D. $O(n \log n)$



Quiz

5. In Big-O notation, which describes the fastest growing function?

A. $O(n)$

B. $O(n \log n)$

C. $O(2^n)$

D. $O(n^2)$

Quiz



6. Which data structure works on the principle of FIFO (First In First Out)?

- A. Stack
- B. Queue
- C. Tree
- D. Graph

Quiz



7. What is the best-case time complexity of Insertion Sort?

A. $O(n)$

B. $O(n \log n)$

C. $O(n^2)$

D. $O(1)$



Quiz

8. The space complexity of an algorithm refers to:

A. Maximum number of instructions executed

B. Amount of memory required

C. Time required for execution

D. Number of variables declared

Quiz



9. A _____ is a Last In First Out (LIFO) data structure.
10. Binary Search requires the input array to be _____.
11. The algorithmic complexity that describes constant time is _____.
12. In Quick Sort, the element used to divide the array is called the _____.



Quiz

```
13. int linearSearch(int arr[], int n, int target) {  
    for (int i = 0; i < (a); i++) {  
        if ((b) == target) {  
            return (c);  
        }  
    }  
    return -1;  
}
```




Quiz 14. `int binarySearch(int arr[], int n, int target) {`
 `int left = 0, right = n - 1;`
 `while ((a) <= (b)) {`
 `int mid = (left + right) / 2;`
 `if (arr[mid] == target) return (c);`
 `else if (arr[mid] < target) left = (d) + 1;`
 `else right = (e) - 1;`
 `}`
 `return -1;}`

Quiz



```
15. void bubbleSort(int arr[], int n) {  
    for (int i = 0; i < (a); i++) {  
        for (int j = 0; j < n - i - 1; j++) {  
            if (arr[j] (b) arr[j + 1]) {  
                (c);  
            }  
        }  
    }  
}
```



Assignment 1

1. Re-implement (C++) learned algorithms:
 - Search: Linear Search, Binary Search.
 - Sorting: Selection Sort, Insertion Sort, Bubble Sort, Quick Sort, Heap Sort, Radix Sort.
2. Re-implement (C++) additional algorithms:
 - Search: Jump Search, Ternary Search, Interpolation Search, Exponential Search.
 - Sorting: Counting Sort, Bucket Sort, Merge Sort, Tim Sort.
3. Report: algorithm idea, characteristics, visualization each step's result (input length < 10).
4. Submit: a `<studentID>.zip` file includes `codes.cpp` and `report.pdf` (should be in latex format).